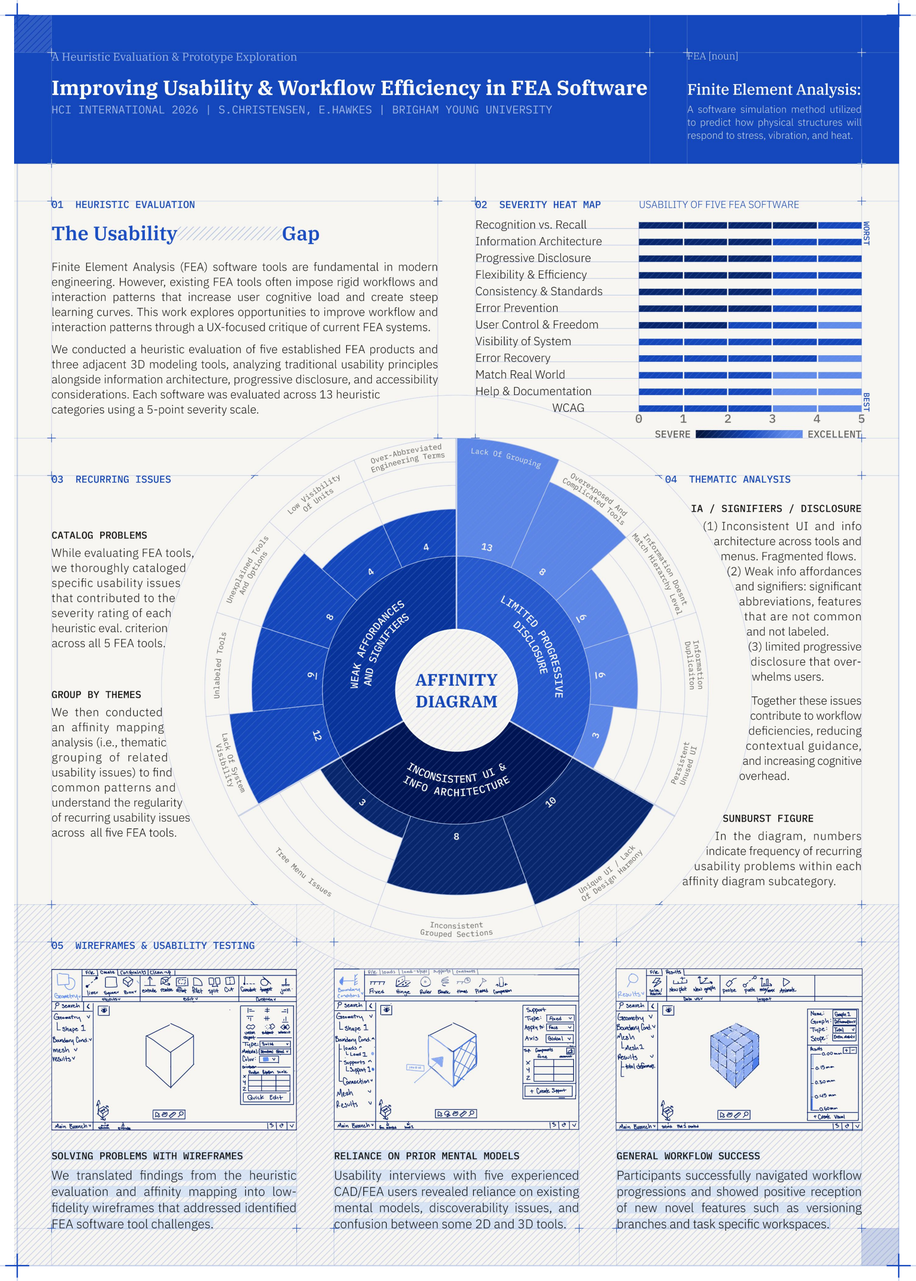
Improving Engineering Software





30 Second Brief

Presented research on engineering software ***in Montreal, CAN at HCII 2026*** with professor Seth Christensen. Led research involving heuristic evaluations and usability testing.

My role

conducted research, synth, and usability tests with proposed design solutions.

Scope

Published paper including research, synthesis and early design solutions.

Impact

Identified 127 issues across industry tool, to inform research driven solutions of future FEA application in ***early development.***

[Full paper ↗]() [Paper poster ↗]()

The Usability Gap

Simulation engineering software is vital to public safety yet is incredibly complex, making it hard to learn, and error prone for even professionals. ***Why?***

Heuristic eval. of top tools

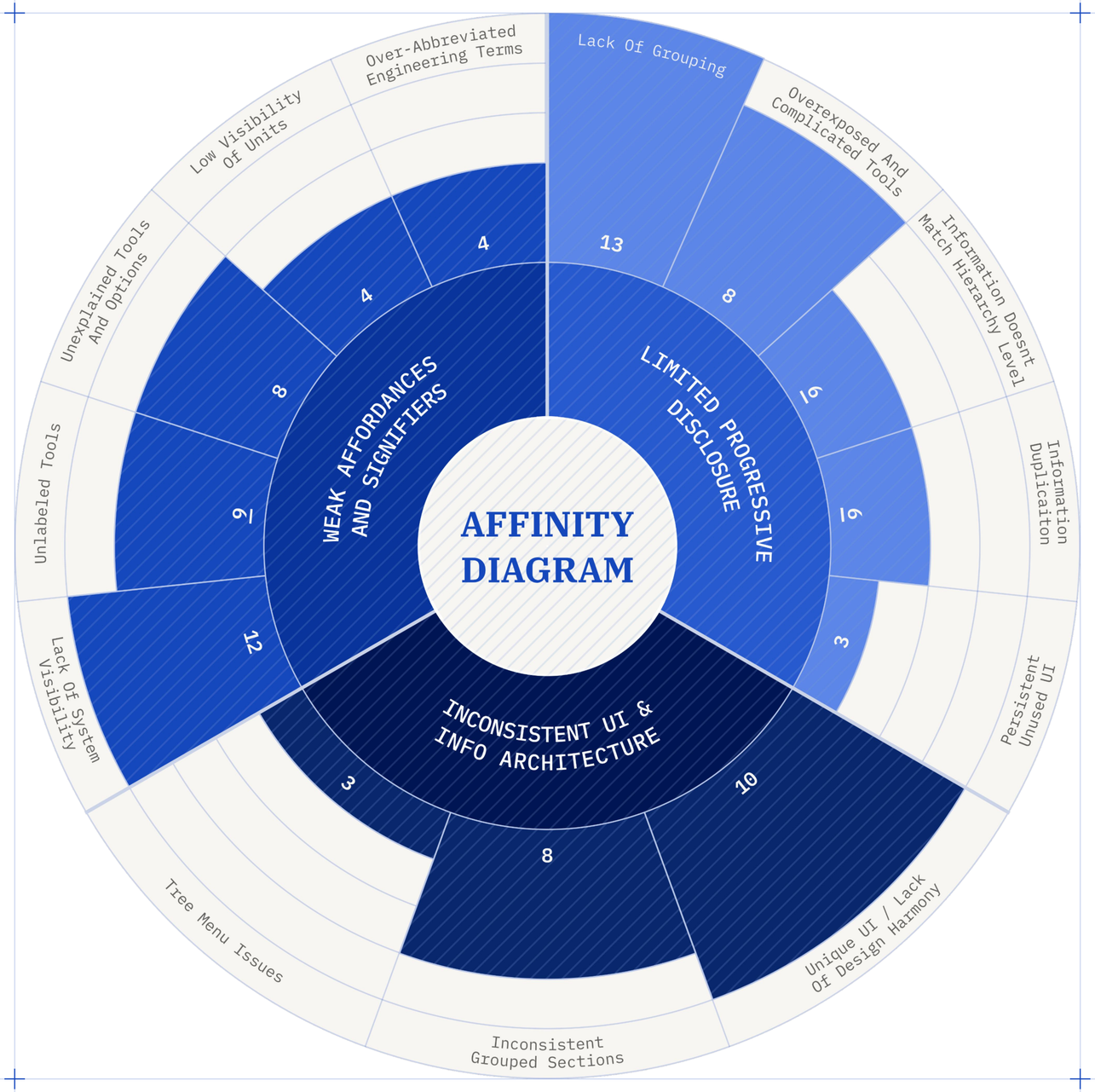
Using common heuristic categories (From NNG) and a few others I cataloged the problems in top simulation tools used in industry.

Most tools did poorly in many categories, with several catalogued problems.

View full heuristic evaluation ↗

Heuristic	Ansys	HyperWorks	NX Siemens	Abaqus	Patron
Visibility of System Status	3	4	3	2	3
Match Between System and the Real World	5	3	3	4	3
User Control and Freedom	4	4	3	3	4
Consistency and Standards	2	3	2	3	2
Error Prevention	2	2	4	2	4
Recognition Rather Than Recall	3	3	4	1	2
Flexibility and Efficiency of Use	3	5	3	2	3
Help Users Recognize, Diagnose, and Recover from Errors	3	2	4	3	N/A
Help and Documentation	4	5	3	N/A	2
Information Architecture (IA)	3	3	2	3	3
Progressive Disclosure	2	3	4	2	3
WCAG Accessibility Standards	5	5	4	4	4
Total Score (out of 65):	39	42	39	29	33

(lower is better)



Three themes emerged

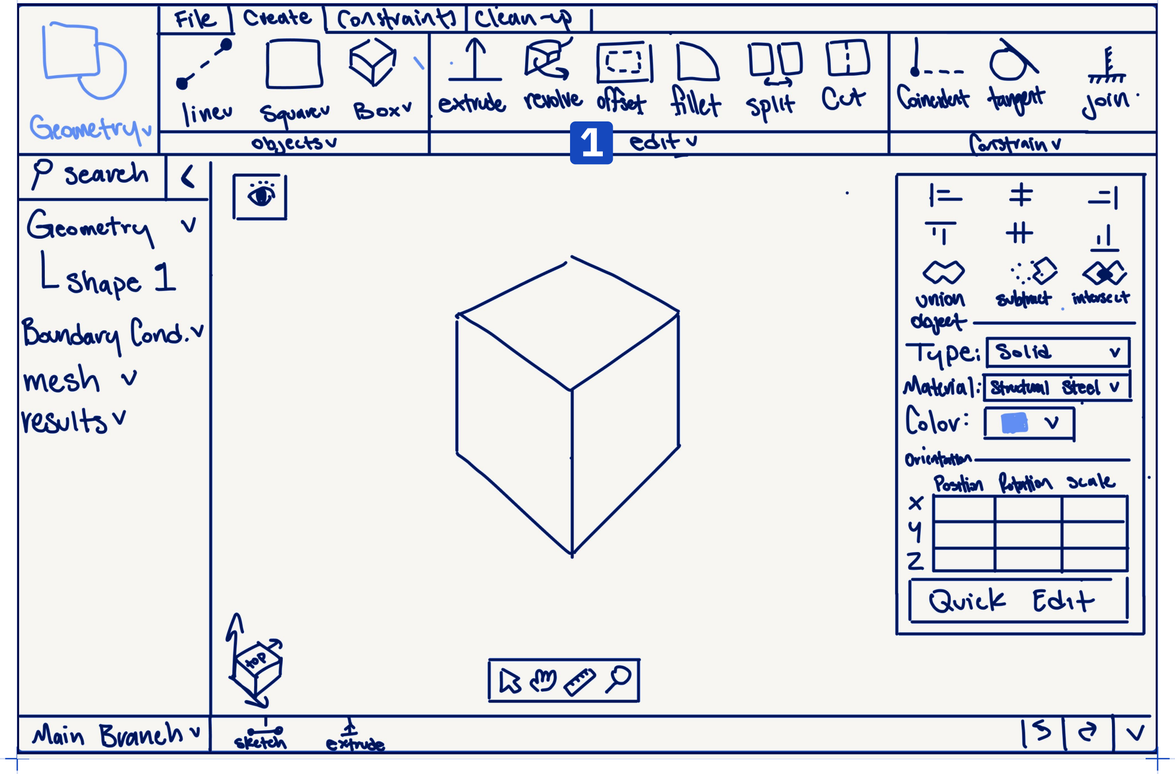
127 total insights from were grouped into categories then placed under larger themes:

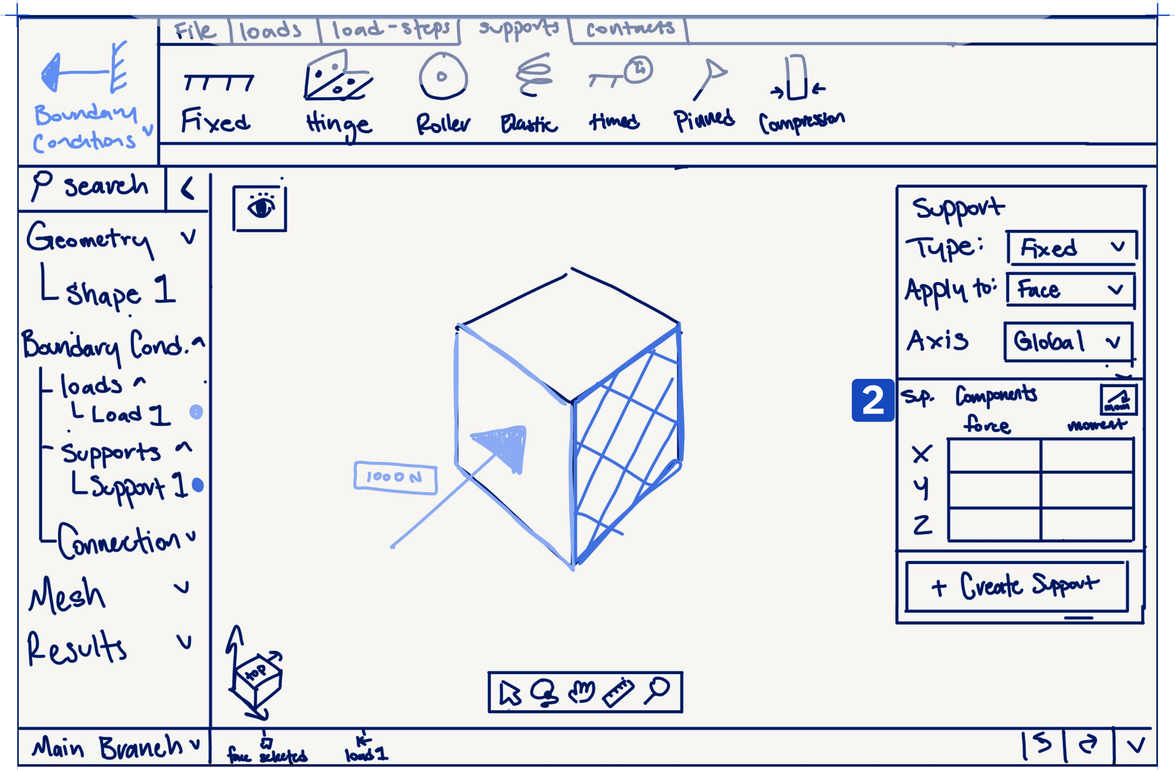
- **Weak Affordances & Signifiers:** lack of tool names, and use of heavy abbreviations
- **Limited Progressive Disclosure:** menus, tools, and settings persistently open.
- **Inconsistent UI & Info Architecture:** lack of system and experience consistency.

Note: Larger sections are broken down into smaller sections with insight counts.

Labels and progressive disc.

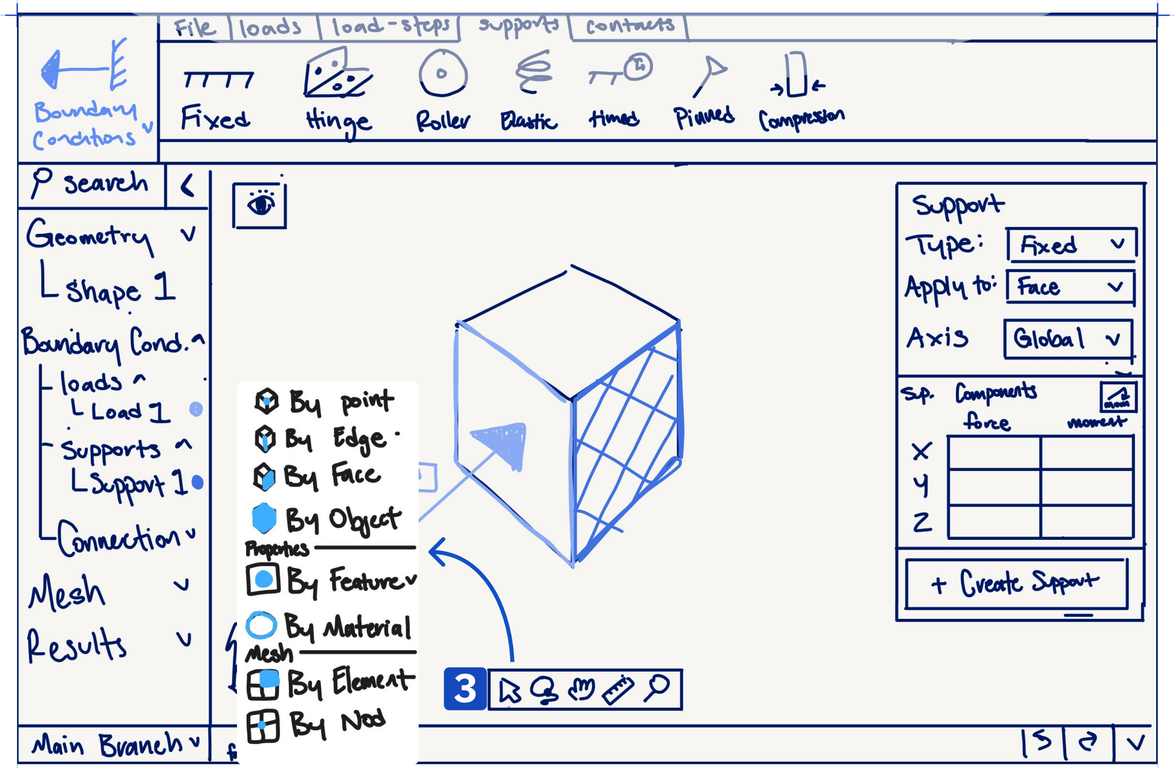
Each tool was given an explicit label and grouped together with similar tools. Less common tools available in the drop down under each section [1].





Contextual menu space [2]

To limit persistently open menus and provide the user with pertinent information the right menu changes depending on what the user is doing.



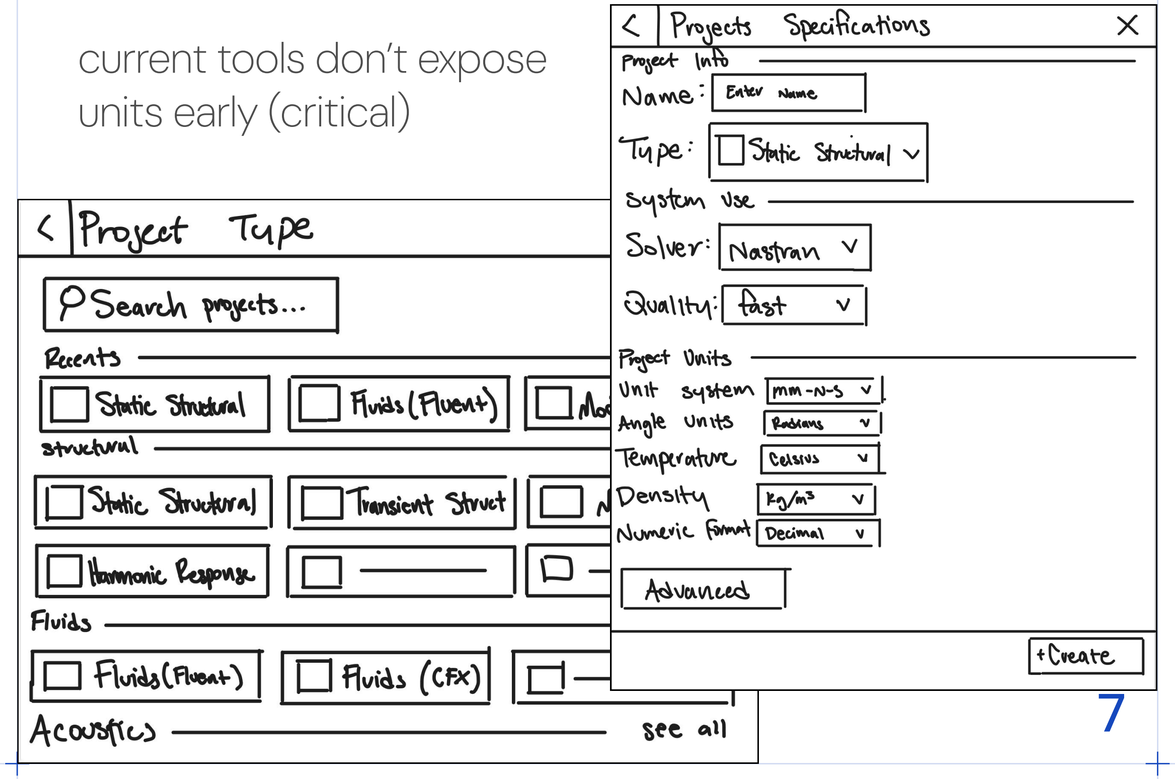
Bottom 3D canvas tools [3]

Tools for canvas navigation are grouped into one area, at the bottom of the canvas, reducing friction for quick 3D space navigation.

Assisted project startup

Different project types are organized by recents, and larger groups (or searchable), and units and project settings are shown early.

current tools don't expose units early (critical)



Insights from 5 user tests

1. Workflow structure was understood and appreciated. (5/5 participants)
2. Existing mental models from using other tools shaped expectations. (4/5)
3. Geometry 2D tools needed better labeling or were missing and needed to be added. (4/5)

Current engineering professors and students completed common tasks with the wireframes

Next Steps

Collaborating with the engineering department, this project will inform the development of a simulation (FEA) tool currently in early stages.

While the proposed solutions provided a clearer workflow, next I would target tools for geometry creation, making sure the follow industry norms.

Exploring Montreal with my professor



Thank You.